ASHUTOSH KUMAR MANDAL

Senior AI Engineer | Applied Generative AI, Agentic Systems & ML Platforms

ashutosh.iitg.16@gmail.com | Bengaluru, India | Portfolio | ashutosh-iitg | ashutosh-iitg

EXPERIENCE

Senior AI Engineer
CombineHealth (UpTrain) — Sep 2024 – Present — Bengaluru, India

Denial Intelligence & Resolution Platform
- Built an end-to-end healthcare revenue-cycle platform that transforms EOB/ERA PDFs into structured claim data, denial analytics, resolution decisions, and appeal packets using multimodal **Claude models on AWS Bedrock** and **Temporal** workflows.
- Engineered a multi-stage extraction pipeline with page-relevance filtering, cached OCR, repeated extraction and consolidation, cross-page claim reconstruction, and payment-model-aware denial logic; explicitly separated patient-liability codes from true denials to preserve downstream metric correctness.
- Architected five durable workflows covering ingestion, enrichment, triage, resolution, and appeal generation. Combined deterministic carve-out rules, configurable business policies, LLM reasoning, and signal-based human approval while keeping PHI in organization-scoped **S3** storage.
- Developed a bronze/silver/gold **ClickHouse** lakehouse using **Vector** and **dbt** with 10 models and 47 data-quality tests. Consolidated two 1,800+ line analyzers into a parameterized query layer and validated migration parity for 3 healthcare accounts through side-by-side testing.
- Built an A/B performance benchmark for production query shapes, including ClickHouse `FINAL` semantics and partition-scoped settings, with a p95 $<$ 50 ms release criterion for migration decisions.
- Designed a statistical denial-driver detection pipeline using two-proportion z-tests and **Benjamini–Hochberg FDR correction** across multi-dimensional claim segments, producing deterministic, statistically supported narratives.

Agentic RCM Research Assistant
- Designed a hierarchical multi-agent pipeline spanning intent classification, complexity-adaptive planning, parallel mini-agent execution, and response synthesis. The planner dynamically routes each query to 1–3 of 9 specialized healthcare agents backed by 27 tools for payer policies, medical coding, NCCI edits, drug information, and general RCM research.
- Implemented two-level asynchronous fan-out using **asyncio**, executing selected mini-agents concurrently while each agent independently plans and invokes its tools in parallel. Built a provider-agnostic, **Pydantic JSON-schema** tool-calling protocol compatible with different LLMs through a LiteLLM proxy.
- Engineered specialized retrieval pipelines for Medicare and commercial payer policies: extracted and chunked policy PDFs, semantically reranked pages using **Jina embeddings**, and combined them with hybrid dense+sparse knowledge-base retrieval using **BGE-M3** and **Milvus**.
- Built context-aware multi-turn interactions that request clarification for ambiguous healthcare queries, maintain persistent session history in **DynamoDB**, and stream responses through **FastAPI** with Redis-based progress signalling.

Automated AR Follow-up
- Developed durable AR follow-up workflows that unified payer integrations across **RPA/Playwright**, APIs, and EDI connectors. Used Temporal timers, webhook signals, pluggable calling providers, and resubmission-aware scheduling to coordinate long-running portal and calling tasks.

AI Engineer
Valuence Technologies — Aug 2020 – Aug 2024 — Tokyo, Japan (Remote)

Conversational Chatbot
- Developed an **agentic RAG** system enabling dynamic question-answering and contextual interactions iterating over proprietary and open-source LLMs like **OpenAI, Mistral** and **Llama**. Integrated support for PDF and spreadsheet documents using **Langchain** and ChromaDB vector storage. Model deployed in production on Azure Cloud as part of Keiko Chatbot.
- Built a RASA-based AI chatbot assistant using transformers for intent classification and optimized its performance by tuning various hyperparameters leading to a 96.8% accuracy in intent classification. Configured bilingual support for English & Japanese language, enhancing system's accessibility and usability. POC deployed using containers on AWS ECS.

Sensitive Data Redaction
- Crafted an innovative pipeline for automated redaction of personally identifiable information in product documents using OCR, multilingual **BERT** based text embedding model and SVM leading to 97.7% accuracy in sensitive data identification.
- Engineered an AI solution to mask sensitive information in product images using **instance segmentation** and image processing. Developed an image processing framework using pixelation to mask the sensitive area such that the edit blends naturally with the original image. Achieved an mAP(50-95) of 0.776 in masking sensitive areas.

Auto-Assessment Platform
- Designed and developed an end-to-end AI workflow for product detection and classification using state-of-the-art **CNN** models. Implemented new architectures like Branched CNN on **Resnet** backbone to identify product details for luxury products.
- Fine-tuned an object detection model using transfer learning to detect target products from images in real-time. Achieved a precision of 97% resulting in a three-fold increase in efficiency and 50% faster response time for customer queries.

PROJECTS

Multi-Modal RAG Document QA — GitHub

- Built a **multi-modal RAG** application for Document question answering for financial documents utilizing Google **VertexAI Embeddings** and **Gemini LLMs** that can understand images, texts and tables. Implemented a persisted vector storage using **ChromaDB** on **LangChain**. Developed the UI with **Chainlit** to allow users to efficiently converse with the application.

Abstract Classification — GitHub

- Fine-tuned **DistilBERT** model using **PyTorch** to each sentence of an abstract into different labels such as Objective, Background, Methods, Conclusions and Research. Achieved an F1 score of 0.92 on test set.

PlantCLEF - Plant Genus and Species Classification — GitHub

- Engineered a Multi-View CNN architecture that integrates multiple images of the same plant to improve classification precision. Achieved a 10% increase in classification accuracy compared to traditional single-view CNN models.

EDUCATION

Indian Institute of Technology Guwahati — Jul 2016 – Jun 2020

Bachelor of Technology in Engineering Physics (7.76/10.0) — Assam, India

TECHNICAL SKILLS

LLM & Agent Systems: RAG, Multi-Agent Orchestration, Tool Calling, Structured Outputs, Prompt Engineering, Evaluation, OpenAI, Claude, Gemini, Azure AI Foundry

LLM Infrastructure: LiteLLM, LLMOps, VLLM, Langchain, Langgraph, Langfuse, Pydantic, BGE-M3, Jina Embeddings, Milvus DB, Hugging Face

Orchestration & APIs: Temporal, FastAPI, Django REST Framework, AsyncIO, Webhooks

Data Systems: ClickHouse, dbt, PostgreSQL, DynamoDB, Redis, S3, Vector Search, Medallion Architecture

Cloud & DevOps: AWS Bedrock, ECS, ECR, EC2, SageMaker, Docker, GitHub Actions, Git

Programming & ML: Python, SQL, C++, PyTorch, Transformers, Scikit-learn, OpenCV